Virulence profiles of non-typhoidal *Salmonella* strains of bovine, porcine and avian origins

Ryan Nguyen[1], Shivdeep Singh[3], Sahar Alshalchi[2], and Sinisa Vidovic[2]

McNair Scholar[1]; Department of Veterinary and Biomedical Sciences[2], Department of Population Medicine[3], Saint Paul, University of Minnesota

Abstract

Non-typhoidal *Salmonella* is the leading zoonotic pathogen in the US**.** This incredible diverse species can infect a wide range of animals and humans. In this research project, we screened a collection of 240 NTS isolates, obtained from several animal species (e.g. avian, bovine and porcine), for the presence of 8 virulence genes (e.g. *siiE*, *sopB*, *bcfC, mgtC, stn, ssaC, invE* and *pefA*). The prevalence of these 8 virulence genes ranges from 31.6% to 97.5%. The gene *pefA* showed the most significant (P<0.000) host-related association, indicating a possible role in the pathogen-host interaction.

Introduction

- *Salmonella enterica* is a rod-shaped Gram-negative bacterium (**Figure 1**) that belongs to the *Salmonella* genus and the *Enterobacteriaceae* family.
- *Salmonella* has 2500 serovars (e.g. different taxonomic groups) based on Kauffmann-White scheme which classification based on a combination of somatic antigens O (e.g. lipopolysccharide) and flagellar antigens H.



Figure 1. *Salmonella* serotype Thypi Illustration, CDC.

- Non-typhoidal *Salmonelle (Salmonella enterica* serovars Enteriditis and Typhimurium) are zoonotic pathogens that can infect either animals or humans, often producing a self-limiting gastroenteritis in their hosts.

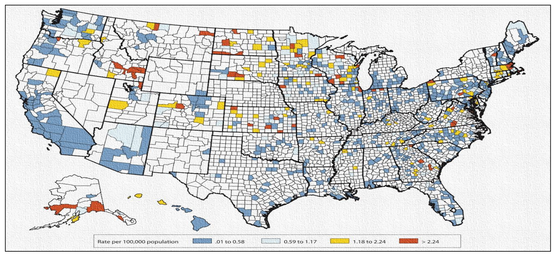


Figure 2. *Salmonella* serotype Enteritidis in 1975. Source: CDC.

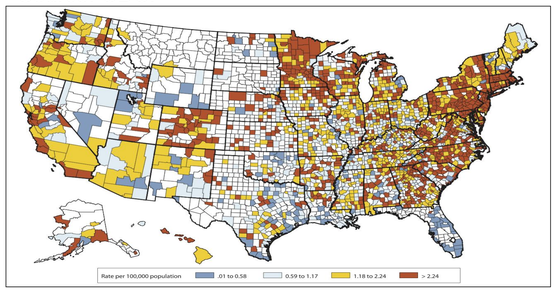


Figure 3. *Salmonella* serotype Enteritidis in 2011. Source: CDC.

- Non-typhoidal *Salmonelle* (NTS) serovars accounted for approximately 1.2 million laboratory-confirmed illnesses, 23,000 hospitalizations, and 450 deaths each year in the United States alone in 2000 to 2008.
- The high incidence rate is made more significant by the fact that there have been no signs of decline in the incidence of salmonellosis over the past 15 years in the United States(**Figures 2 and 3**).
- In this study, we created a large collection of NTS strains (n=240) obtained from three major food-producing animal species: avian (n=80), bovine (n=80) and porcine (n=80). The entire collection of NTS strains have been screened for 8 virulence genes in order to find their virulence potency and possible the host specific-required virulence factor, necessary for successful invasion.

Methodology

Table 1. The primers sequences and expected PCR size of 8 virulence factors used in this project.

Name and symbol of the genes	Primer sequence	Expected PCR size
Fimbrial ushur ***(bcfC)***	Forward primer: **CCAGTACGCTGGCGGATAAT**; Reverse primer: **TGTCATCGTCATAGCCGCTC**	177 bp
Protein MgtC (***mgtC***)	Forward primer: **CCAGTACGCTGGCGGATAAT**; Reverse primer: **TGTCATCGTCATAGCCGCTC**	458 bp
Secretion system apparatus outer membrane protein ***(ssaC)***	Forward primer: **ACCTGGTTTGATGGCAGCAT;** Reverse primer **CCACTAGCACCACCGTCATT**	680 bp
Invasion Protein ***(invE)***	Forward primer: **TCCAGTCGACGGACGAAATG**; Reverse primer: **TAGTACGACGCTGTTCTGCC**	948 bp
Plasmid-encoded major fimbrial subunit ***(pefA)***	Forward primer: **TCCAGTCGACGGACGAAATG**; Reverse primer : **TAGTACGACGCTGTTCTGCC**	165 bp
Enterotoxin ***(stn)***	Forward primer: **CCGCGCCTTTACCCTCAATA** Reverse primer: **CAGGATGCCCAAAGCAGAGA**	361 bp
Inositol phosphate phosphatase ***(sopB)***	Forward primer: **TTGTGGATGTCCACGGTGAG** Reverse primer: **TTATAGGGTTCGCCGCCATC**	644 bp
Hypothetical protein ***(siiE)***	Forward primer: **CAGCGAGGAGAGCGATAGTG** Reverse primer: **ATACTACGCCCCAGTGTCCT**	443 bp

Extraction of DNA

- *Salmonella* strains were plated from frozen stocks on LB agar plates and incubated overnight at 37ºC.
- Bacterial growth from approximately one quarter of the plate containing isolated colonies was collected using a sterile cotton swab and suspended in 1 mL of 0.9% saline.
- After centrifugation at 10000 X g for 1 min the supernatant was removed, and genomic DNA was extracted by using the Qiagen Dneasy.

Virulence Gene Profiling

- Duplex PCR reactions were used for the detection of *mgtC and bcfC, pefA and siiE, stn and sopB*. Identification of remaining two virulence genes, *ssaC and invE* were performed individually by employing conventional PCR methods.
- The positive control is *Salmonella enterica* ATCC 4931, and the negative control is *Escherichia coli* O157 B-1.
- PCR was carried out in 50 µl reaction volumes using an Bio Rad T100.

Results

Table 2. Percentage of virulence genes in three major animal food-producing species (red is bovine, green is porcine, and blue is avian)

	Prevalence of virulence genes (%, n=240)								
	No. of isolates	*pefA*	*siiE*	*stn*	*sopB*	*bcfC*	*mgtC*	*ssaC*	*invE*
Dublin	50	0	100 (n=50)	100 (n=50)	90 (n=45)	96 (n=48)	100 (n=50)	76 (n=38)	96(n=48)
Cerro	16	0	100 (n=16)	100 (n=16)	81.2 (n=13)	75 (n=12)	75 (n=12)	87.5 (n=14)	81.25 (n=13)
Montevedio	14	0	100 (n=14)	100 (n=14)	92.8 (n=13)	100 (n=14)	100 (n=14)	92.86 (n=13)	92.86 (n=13)
4,5,12:I:-	40	50 (n=20)	100 (n=40)	100 (n=40)	100 (n=40)	90 (n=36)	85 (n=34)	82.5 (n=33)	100 (n=40)
Typhimurium	40	62.5 (n=25)	97.5 (n=39)	100 (n=40)	97.5 (n=39)	92.5 (n=37)	90 (n=36)	92.5 (n=37)	75 (n=30)
Heidelberg	80	38.7 (n=31)	88.7 (n=71)	92.5 (n=74)	80 (n=64)	58.75 (n=47)	67.5 (n=54)	88.75 (n=71)	80 (n=64)
Total	240	31.6 %	95.8 %	97.5 %	89.1 %	80.83 %	83.33 %	85.83%	86.67%
***P* value(≤0.05)**		0.00	0.008	0.069	0.003	0.000	0.000	0.223	0.002

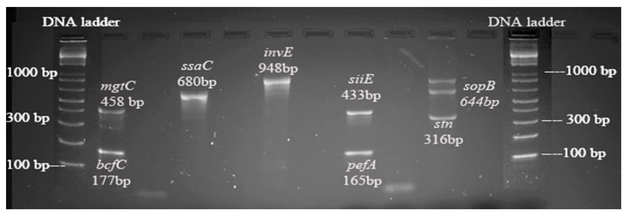


Figure 4: The electrophoresis image of 8 virulence genes factors

- Out of 8 virulence genes, the lowest prevalence rate exhibited the *pefA* gene with being detected in 31.6% of tested NTS isolates, whereas the *stn* gene was the most common with the prevalence rate of 97.5%.
- The most significant host-related association (P<0.000) showed the *pefA* gene.

Discussion and Conclusion

- This study showed that NTS population obtained from three major animal-food producing species (e.g. avain, bovine and porcine) carry high percentage of virulence genes. The *pefA* gene showed overall a low prevalence rate and most importantly this virulence gene have had the most significant association with the animal host.
- The *pefA* gene could play an important role in colonization of certain animal species, indicating that this gene could be a possible target for vaccine development.